

Scalability & Future Plan

Date	03 July 2026
Team ID	XXXXXX
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses limited input parameters	May reduce prediction accuracy	High
2	No real-time weather data integration	Requires manual data entry	High
3	No user login or data history	Previous predictions cannot be saved	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of users	Deploy on a cloud platform to support more users
2	Data Storage	No database integration	Add a database (MySQL/Firebase) to store prediction history
3	Performance	Suitable for a small dataset	Optimize the ML model and use larger datasets for better accuracy
4	Security	Basic application security	Add user authentication and secure data handling

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Integrate real-time weather API	1 Month	More accurate predictions
Phase 2	Add GPS/location-based flood prediction	2 Months	Location-specific alerts
Phase 3	Develop an Android mobile application	3 Months	Easier access for users
Phase 4	Deploy on cloud with SMS/Email alert system	4 Months	Wider accessibility and instant flood notifications